

GraphLIME on RDF knowledge graphs

Explaining an R-GCN in the vocabulary of the graph it was
trained on

01 — CONTEXT

RDF knowledge graphs

RDF triple store

```
person42  works_at  acme
person42  author    paper7
```

A node is an identifier and nothing else. Everything it “is” sits on its typed edges.

Labeled property graph

```
(:Person {name, age, city})
-[:WORKS_AT]-> (:Company)
```

Nodes carry key–value properties. A feature vector already exists.

I started from a GraphLIME written for the graph on the right. My data was the graph on the left.

01 — CONTEXT

R-GCN: the classifier I explain

$$h_i^{(l+1)} = \sigma \left(\sum_{r \in \mathcal{R}} \sum_{j \in \mathcal{N}_i^r} \frac{1}{c_{i,r}} W_r^{(l)} h_j^{(l)} + W_0^{(l)} h_i^{(l)} \right)$$

ONE WEIGHT MATRIX PER RELATION

In a knowledge graph the meaning is in the edge type. A plain GCN would treat *author* and *employs* as the same neighbour.

BASIS DECOMPOSITION

45 relations on AIFB means 45 matrices. Each W_r is a combination of 30 shared bases instead.

02 — THE PROBLEM

The explanation I wanted

WHAT AN R-GCN ON ONE-HOT IDS CAN SAY

```
entity #4211 mattered
```

WHAT MY PIPELINE PRINTS

```
 $\beta=0.37$  in:swrc:carriesOut
```

A domain expert can check the second line against the graph. That is the whole point of the project.

Why GraphLIME does not fit RDF

GraphLIME reads a node feature matrix and asks, through kernels, which columns the model's output depends on. An RDF entity has no columns. There is nothing to select from.

So the first thing I had to do was not to run the method. It was to decide what its input should be.

03 — MY DESIGN

Building a feature space: two attempts

SAME THREE TRIPLES

```
p1 author paper7 · p1 author paper9 · p1 worksAt aifb
```

Space A — predicate counts

```
row p1 → out:author = 2  
         out:worksAt = 1
```

One column per predicate and direction. Coarse and dense: about 90 columns on AIFB.

Space B — predicate-object pairs

```
row p1 → out:author=paper7 = 1  
         out:author=paper9 = 1  
         out:worksAt=aifb = 1
```

One column per (predicate, term). It can say *author of paper7*, so the columns multiply and the rows go sparse.

Either way X is $N \times C$, one row per entity, and the R-GCN consumes it in place of the one-hot ids. Model and explainer then speak one vocabulary, which is also what makes fidelity measurable.

Why space B failed

Not because of the number of columns, which is the answer I expected. The kernels were the problem: on a matrix that sparse there was almost no signal to measure, and HSIC Lasso could not separate anything.

I ran the rest of the experiments on space A, which also works on more graphs. Space B stays in as a comparison.

HSIC: what I use it for

$$\widehat{\text{HSIC}}(K, L) = \frac{\text{tr}(KHLH)}{(n-1)^2}$$

RBF kernels, median-heuristic bandwidth. With characteristic kernels it is zero exactly when the two variables are independent.

HSIC is good at spotting **non-linear** correlations between two variables.

A GNN output is not a linear function of its input, so a linear surrogate would measure the wrong thing.

I use HSIC as a tool. I stopped at what it measures and did not go into estimator theory — the interpretation is what I need to defend a ranking.

04 — METHOD

HSIC Lasso in three terms

$$\min_{\beta \geq 0} \underbrace{- \sum_k \beta_k \text{HSIC}(x_k, y)}_{\text{relevance}} + \underbrace{\frac{1}{2} \sum_{k,l} \beta_k \beta_l \text{HSIC}(x_k, x_l)}_{\text{redundancy}} + \underbrace{\rho \|\beta\|_1}_{\text{sparsity}}$$

Relevance

Rewards features the output actually depends on.

Redundancy

Penalises features that repeat each other. A duplicated feature splits its weight.

Sparsity

Forces most weights to zero, so an explanation stays short.

04 — METHOD

Pipeline and data preparation

01

Parse RDF into a deterministic multigraph

02

Build the vocabulary and the matrix X

03

Train the R-GCN on X , not on ids

04

Explain: 2-hop neighbourhood, HSIC Lasso

05

Evaluate fidelity, stability, agreement

WHERE THE TIME WENT

Stripping the predicates that make the label obvious, and folding raw RDF text into count vectors. A lot of regex.

WRITTEN FROM SCRATCH

HSIC and the solver, with property tests written before the implementation: symmetry, nHSIC in $[0,1]$, weight splitting, sparsity in ρ .

Datasets, model, settings

Dataset	Nodes	Triples	Predicates	Task
AIFB	8,285	29,043	45	Research-group affiliation, 4 classes
MUTAG	23,644	74,227	23	Mutagenicity, 2 classes

MODEL

Two RGCNConv layers, 30 bases, hidden 32. Adam, lr 0.01, 200 epochs, 5 seeds.

EXPLANATIONS

2-hop neighbourhoods, RBF kernels, $\rho = 0.1$. Under 10 neighbours the pipeline refuses.

05 — SETUP

A ground-truth check before real data

On a generated graph, class 1 holds if and only if the entity has predicate p7.

I know the answer, so I can check the explainer instead of trusting it.

$\geq 95\%$

of explained nodes ranked
out : syn : p7 first at zero noise

top-3

even with two distractor predicates
correlated 90% with the target

first run

it passed, and no threshold was lowered
afterwards to keep it passing

How I measure an explanation

Fidelity+ — necessity

Zero the top-k columns. If they matter, the predicted probability falls. Higher is better.

Random-k control

Masking any k columns perturbs the model. The evidence is the gap against random, not the raw value.

Fidelity- — sufficiency

Keep only the top-k. If they are enough, the prediction survives. Lower is better.

Stability — Jaccard@5

I explain the same entity twice under a condition that should not matter — a different training seed, or a different radius — and compare the two top-5 predicate lists: shared predicates divided by distinct predicates. Identical lists give 1, no overlap gives 0. Averaged over every explained entity.

Refusal rate

GraphLIME's samples are the target's neighbours, and HSIC estimated from a handful of samples is noise. Below 10 neighbours I return no explanation and count the entity as refused. At 2 hops the rate was 0 of 36 on AIFB and 0 of 68 on MUTAG: the guard was never needed at

06 — RESULTS

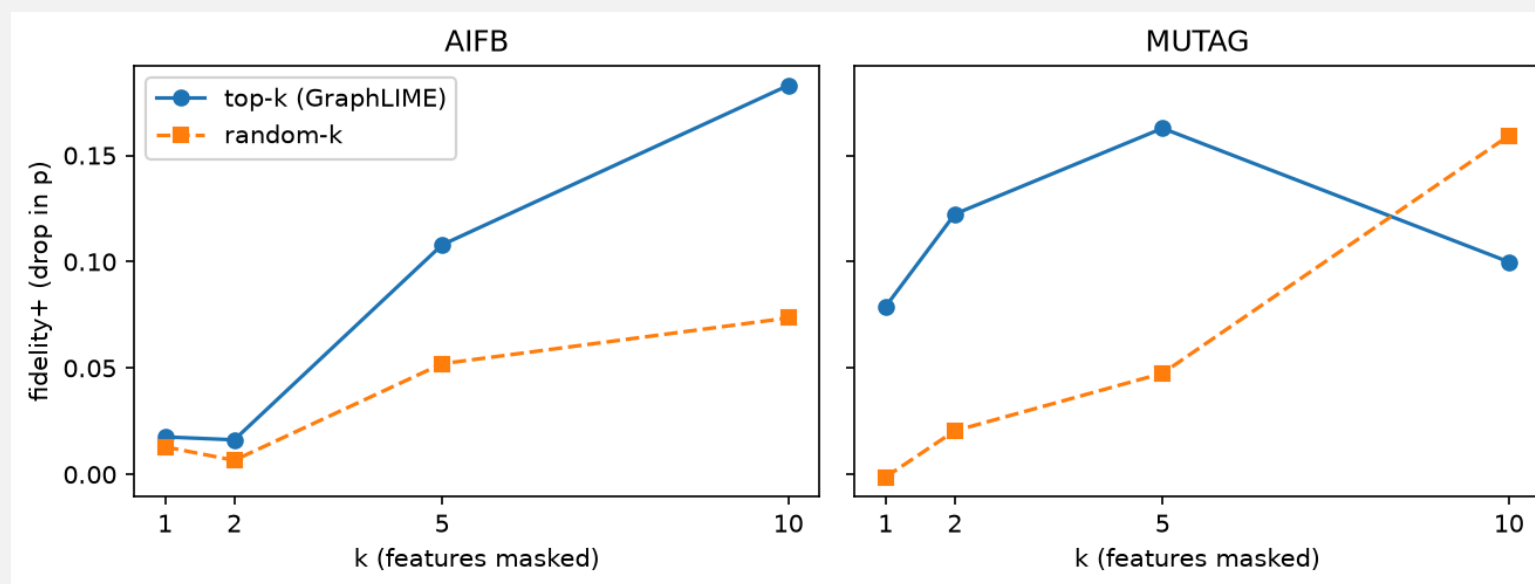
Classification accuracy

Dataset	Test accuracy, 5 seeds	Best seed	R-GCN paper, one-hot
AIFB	0.8833 $\pm$ 0.0534	0.9444	0.958
MUTAG	0.7559 $\pm$ 0.0305	0.7941	0.732

AIFB classes depend on *which* entity a person is connected to, and predicate counts throw that away on purpose. I pay for readability in the input representation, not in the explainer.

06 — RESULTS

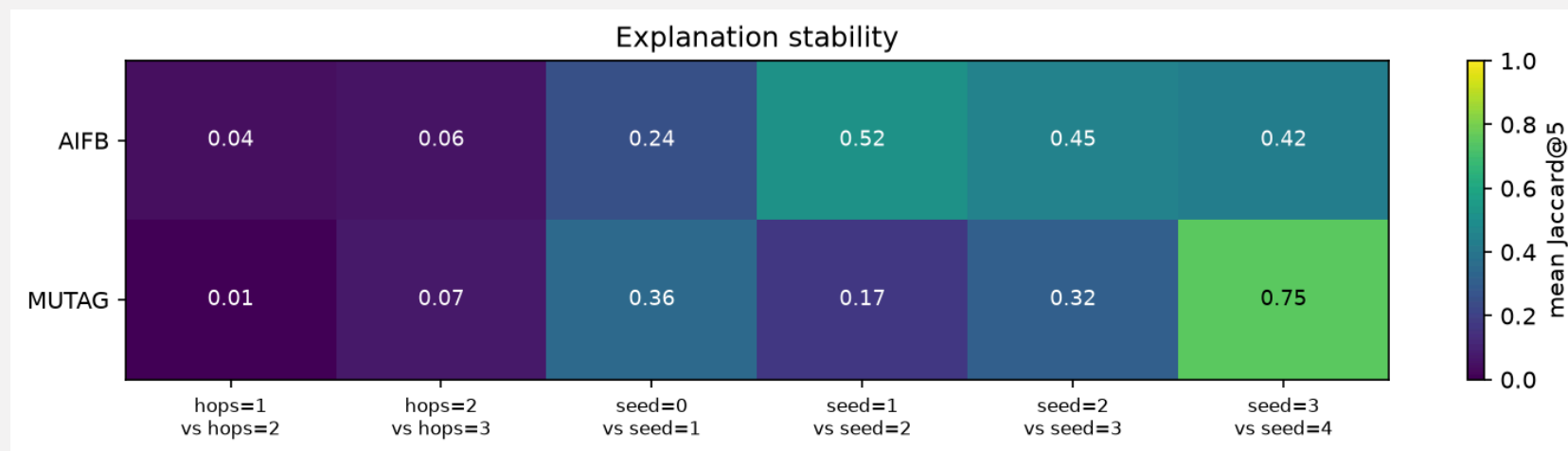
Fidelity+ against the random-k control



Up to $k = 5$ the explanations beat the control on both datasets. The inversion at $k = 10$ on MUTAG comes from the sweep: my explanations carry fewer than ten non-zero β , so the top-10 set pads itself with irrelevant columns.

06 — RESULTS

Stability across seeds and hops



Across seeds, 0.17 to 0.75. Across hops, 0.01 to 0.07. GraphLIME's samples *are* the neighbourhood, so an explanation is only well defined once the radius is fixed and stated.

Feature space and baseline agreement

Comparison	Mean Jaccard@5 over predicates
AIFB — space A vs space B	0.2843
MUTAG — space A vs space B	0.0029
AIFB — GraphLIME vs GNNExplainer	0.1456
MUTAG — GraphLIME vs GNNExplainer	0.3153

The choice of vocabulary changes the answer, and on MUTAG it changes it completely. GNNExplainer’s edge mask cannot run on RGCNConv, so the comparison covers feature attributions only.

07 — CLOSING

Limitations

- 01** β is dependence, not causation, and it is local to one neighbourhood.
- 02** Fidelity is measured on columns. Masking one removes the evidence of a predicate but leaves the edges in place.
- 03** Subsampled above 200 neighbours, refused below 10, and in between the radius dominates.
- 04** 176 and 340 labelled entities. Every mean rests on a few tens of test nodes.
- 05** The baseline is not like-for-like: GNNExplainer is compared through its attribute mask.

07 — CLOSING

What I take from the project

01

On RDF the work is not running GraphLIME. It is deciding what its input should be.

03

The explanations beat the random control up to $k = 5$, and recovered a planted rule on synthetic data.

02

Letting the model consume the same matrix is what makes fidelity measurable at all.

04

At this graph size the neighbourhood radius dictates the explanation. That bounds how far one of them should be trusted.